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Microstructure Regulation and Optoelectronic Performance Optimization of Flexible CPI-Based ITO Thin Films Under Low-Temperature Heat Treatment Process

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Abstract

Addressing the urgent need for low-temperature processes in the manufacturing of flexible vehicle-mounted touch display devices, this study investigates the process–structure–performance relationships of indium tin oxide (ITO) thin films prepared by DC magnetron sputtering on transparent polyimide (CPI) substrates. A synergistic strategy of "low-temperature deposition (110 °C)–230 °C atmospheric annealing" was employed. The optimal sample exhibited excellent comprehensive performance: a resistivity as low as 203 μΩ·cm, an average visible light transmittance of 89.2%, a surface roughness of 0.76 nm, and the ability to endure 100,000 bending cycles at a radius of R = 5 mm with a sheet resistance change rate of less than 10%. Microstructural and chemical state analyses revealed that this process facilitates the complete oxidation of Sn^{2+} to Sn^{4+} (Sn^{4+}/Sn^{2+} ratio of 8.2:1) and the controlled formation of oxygen vacancies (O_L/O_V ratio of 6.5:1), leading to a synergistic improvement in carrier concentration (8.7×10^{20} cm^{-3}) and mobility (35.2 $cm^2/V{\cdot}s$). This work elucidates the crystallization kinetics and doping mechanisms under low-temperature conditions, providing a viable low-temperature technical pathway for the fabrication of high-performance transparent electrodes in flexible electronics.

Keywords: ITO thin films; flexible vehicle-mounted touch display devices; low-temperature process; microstructure regulation; optoelectronic properties; transparent polyimide (CPI)

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1. Introduction

The rapid advancement of flexible electronics has significantly increased the demand for high-performance flexible materials. Emerging applications such as wearable health monitors, flexible robotic skin, implantable medical electronics, and vehicle-mounted flexible touch displays [1–4] require electronic components to maintain excellent electrical performance and reliability under various bending and folding deformations. This trend poses a fundamental challenge to traditional rigid electronic devices and has spurred extensive research on flexible transparent conductive materials.

Among various transparent conductive materials, indium tin oxide (ITO) has long dominated the market due to its excellent optoelectronic properties and mature preparation processes [5]. However, its application in flexible devices faces significant challenges.

Commonly used flexible plastic substrates, such as polyethylene terephthalate (PET) and polyethylene naphthalate (PEN), have low glass transition temperatures (Tg) and cannot withstand the high deposition temperatures and post-annealing treatments of traditional ITO processes [6]. ITO films prepared at low temperatures often suffer from poor crystalline quality and high defect density, resulting in inferior electrical properties compared to films prepared on glass substrates [7,8]. Furthermore, the significant coefficient of thermal expansion mismatch between PET/PEN and ITO introduces substantial internal stress during device bending, leading to film cracking or delamination, which severely limits mechanical reliability and service life [9].

Transparent polyimide (CPI) offers an ideal solution to the above bottlenecks. CPI possesses excellent mechanical properties, chemical stability, high-temperature resistance (Tg > 350 °C), and its thermal expansion coefficient can be tailored through molecular design to better match ITO films. Although CPI has many advantages, developing low-temperature processes for high-performance ITO films remains crucial for compatibility with heat-sensitive organic functional layers in flexible electronics manufacturing [10,11].

Recently, various nanomaterials have been investigated as potential alternatives or complements to ITO for flexible optoelectronic devices. Carbon nanostructures, particularly graphene and carbon nanotubes, show great promise due to their outstanding mechanical flexibility, high conductivity, and visible light transparency [12–14]. In addition, the possibility of preparing solution-processable electrodes based on carbon nanomaterials enables large-area deposition and reduces the cost of fabrication [12]. Similarly, Carbon nanotubes (CNTs) and graphene (Gr) are promising materials for flexible transparent conductive films, exhibited an excellent optoelectronic performance (Rs = 45.6 Ω/sq., T = ca. 80% at 550 nm) [13]. While these emerging materials offer new solutions, ITO remains irreplaceable in demanding applications like vehicle displays due to its long-accumulated process maturity, excellent uniformity, and environmental stability.

Although studies exist on low-temperature preparation of ITO films on CPI substrates, our team's previous research focused on improving the bending resistance of CPI-based ITO films, emphasizing structural optimization [15]. This study systematically investigates the synergistic effect of "deposition temperature–post-annealing treatment" and provides an in-depth explanation of its microscopic mechanism. The scientific question of how the deposition temperature affects the film's microscopic crystal structure, thereby determining the effectiveness of subsequent low-temperature annealing, has not been clearly addressed.

Based on this, we innovatively propose a "low-temperature deposition–post-annealing synergistic regulation" strategy, aiming to prepare high-performance ITO films at temperatures far below the CPI tolerance limit. This paper systematically studies the regulation of substrate temperature (30–300 °C) during DC magnetron sputtering on the microstructure and chemical state of ITO films, and combined with 230 °C atmospheric annealing, analyzes its impact on the final optoelectronic properties and surface morphology. Using characterization methods such as XRD, SEM, AFM, and XPS, we reveal the Sn^{2+} to Sn^{4+} conversion, oxygen vacancy formation, and crystallization kinetics under low-temperature conditions, clarifying the physical essence of the synergistic optimization effect.

2. Experimental Methods

2.1. CPI Substrate and ITO Film Preparation

CPI films were prepared on glass carrier plates using the slot-die coating method. Soda-lime glass substrates were ultrasonically cleaned with a 5% NaOH solution and deionized water, dried with N_2, and then CPI slurry was uniformly coated using a slot-die head at 5 mm/s. The coated samples were heat-treated in an atmospheric environment at

350 °C for 1 h to fully imidize and cure, obtaining free-standing flexible CPI films with a thickness of 10 ± 0.5 um (Figure 1).

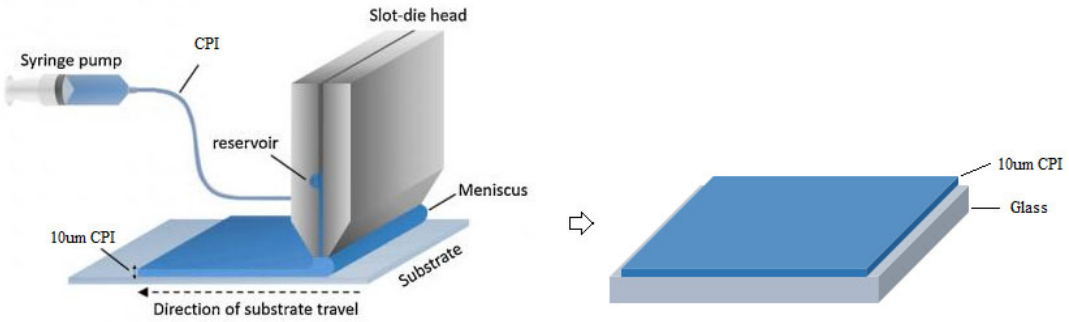


Figure 1. Fabrication process of 10 um-thick flexible CPI substrates.

ITO films were deposited using a Leybold Optics Inline A1100V4 DC magnetron sputtering system (Leybold Optics, Cologne, Germany) with a ceramic target (99.99% purity) of In_2O_3:SnO_2 = 90:10 wt%. (Figure 2). The key deposition parameters are summarized in Table 1. The film thickness was precisely controlled at 30 ± 2 nm, as confirmed by a surface profilometer (Dektak XT, Bruker). A series of samples were prepared within the substrate temperature range of 30–300 °C.



Figure 2. Leybold Optics Inline A1100V4 DC magnetron sputtering system.

Table 1. Parameters for ITO deposition by DC magnetron sputtering.

Parameter	Value
Base Vacuum	1.0×10^{-4} Pa
Process Pressure	0.5 Pa
Power Density	0.67 W/cm^2
Substrate Temperature	30–300 °C
Ar Flow Rate	120 sccm
Deposition Time	Controlled for 30 ± 2 nm thickness

2.2. Heat Treatment Process

A atmospheric annealing furnace was used for post-annealing treatment of the samples. The deposited samples were heated from room temperature to 230 °C at a rate of 5 °C/min,

held at that temperature in an oxygen atmosphere (flow rate 200 sccm) for 60 min, and then naturally cooled to room temperature.

2.3. Performance Characterization

X-ray diffraction (XRD, Shimadzu 7000, Shimadzu Corporation, Kyoto, Japan) was used to analyze the film crystal structure; atomic force microscopy (AFM, Multimode Nanoscope IIIa, Bruker, Billerica, MA, USA) was used to characterize surface morphology and roughness; field emission scanning electron microscopy (FESEM, Zeiss Gemini 300, Carl Zeiss AG, Oberkochen, Germany) was used to observe the microstructure; a four-point probe tester (SDY-4, Guangzhou Four Probes Technology Co., Ltd., Guangzhou, China) was used to measure resistivity; a Hall effect measurement system (HMS-3000, Ecopia Corp., Anyang, Republic of Korea)was used to measure carrier concentration and mobility; UV-Vis (UV-2600, Shimadzu Corporation, Kyoto, Japan)spectrophotometry was used to measure optical transmittance; X-ray photoelectron spectroscopy (XPS, PHI-1800, ULVAC-PHI, Chigasaki, Kanagawa, Japan) was used to analyze the chemical state of elements.

2.4. Bending Performance Test

Stress was applied to the sample through a fixture to bend it into an arc shape, treating the bent part as an approximate arc, which is the bending radius of the sample. Combined with the design requirements of touch screen products, this study adopted inward bending, meaning the formed arc has CPI on the outside and ITO on the inside. When adjusting the bending radius, one end of the sample was fixed, and the other end was bent with the movement of the fixture. The bending test setup is shown in Figure 3.

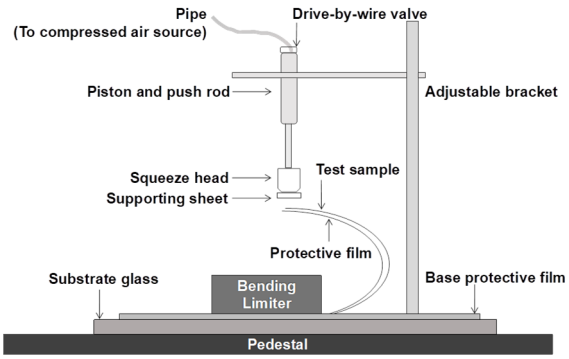


Figure 3. Schematic diagram of a bending testing machine.

3. Results and Discussion

3.1. Structural Properties

Crystal Structure and Microstructure Evolution

XRD tests in Figure 4a show that regardless of deposition at substrate temperatures of 110 °C, 150 °C, 190 °C, or 230 °C, the ITO films still exhibit the cubic bixbyite structure of In_2O_3, and no diffraction peaks of SnO or SnO_2 appear, meaning Sn is incorporated into the crystal structure of In_2O_3. The crystal structure of the ITO films is polycrystalline. When the substrate temperature gradually increases from 110 °C to above 190 °C, the preferred crystal plane of the ITO films gradually changes from (222) to (400), while the diffraction peak intensities of the (222), (440), and (622) crystal planes gradually increase. Figure 4b shows that as the substrate temperature increases, the peak intensity ratio of (222) to (440) and (400) gradually decreases, while the peak intensity ratio of (440) to (400) first decreases and then increases, indicating that ITO crystals gradually grow in the <100> direction.

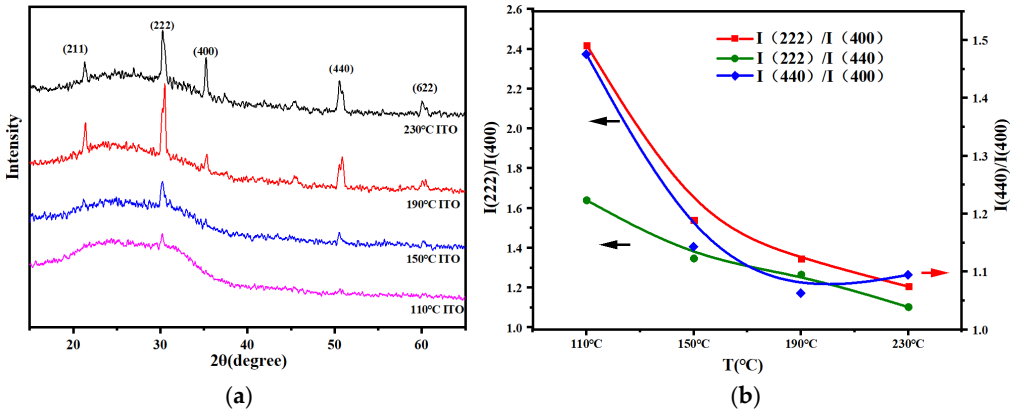


Figure 4. (**a**) XRD patterns of ITO films deposited at different substrate temperatures and (**b**) variation in ITO peak intensity ratio.

According to the Debye–Scherrer formula $D = 0.9\lambda/\beta\cos\theta$, where λ is the X-ray wavelength (0.154 nm), β is the full width at half maximum (FWHM) of the diffraction peak [16], and the lattice constant calculation formula is $1/d^2 = (h^2 + k^2 + l^2)/a_0{}^2$, where (h,k,l) are the Miller indices; the microstructure parameters of the samples obtained from Figure 4a are shown in Table 2:

Table 2. Microstructure of unannealed samples at different substrate temperatures.

Substrate Temperature	(222) Grain Size (nm)	(222) d-Spacing (nm)	(400) d-Spacing (nm)	Lattice Constant a (nm)
110 °C	5.41	0.2929	0.2537	1.01468
150 °C	10.13	0.2926	0.2534	1.01347
190 °C	18.97	0.2921	0.2530	1.01196
230 °C	20.48	0.2921	0.2530	1.01193

Note: Calculations for grain size examples provided in original notes: *1 = 0.9 × 0.154/0.861629 × 0.26 × 0.01745*; *2 = 0.9 × 0.154/0.861629 × 0.45 × 0.01745*; *3 = 0.9 × 0.154/0.861629 × 0.84 × 0.01745*; *4 = 0.9 × 0.154/0.861629 × 0.91 × 0.01745*.

In ITO films, the radii of Sn^{4+}, Sn^{2+}, and In^{3+} ions are 0.69 Å, 0.93 Å, and 0.79 Å, respectively. From Table 2, it can be seen that as the substrate temperature before annealing increases, the lattice constant gradually decreases, and the grain size gradually increases. Since no oxygen was added during the experiment, the Sn in the ITO films was not fully oxidized, partially existing in the Sn^{2+} state. Due to the larger ionic radius of Sn^{2+}, the lattice constant increases. Under oxygen-deficient conditions, even when the temperature reaches Ts, the lattice constant of the ITO film is still larger than the standard In_2O_3 value of 1.0118 nm. Therefore, insufficiently oxidized Sn^{2+} ions may exist in the ITO films. Y. Shigesato et al. proposed the Volmer–Weber model for ITO film growth, where ITO films grow gradually from small islands, and island density is affected by grain aggregation [10]. Before annealing, during ITO film sputtering, particles at higher substrate temperatures gain greater free energy, and their migration ability increases. In contrast, the migration ability of particles at lower substrate temperatures is reduced, forming fine islands. Therefore, the grain radius is smaller, the island density is higher, and grain boundaries increase. The movement of high-energy particles also leads to an increase in (400) plane intensity. Consequently, the resistivity of ITO before annealing decreases with increasing substrate temperature.

ITO films prepared at different substrate temperatures were annealed in an air environment at 230 °C for 1 h. The XRD structure after annealing is shown in Figure 8, and the grain size and lattice constant calculated according to the Scherrer formula are shown in Table 3.

Table 3. Microstructures of samples annealed at 230 °C under different substrate temperatures.

Substrate Temperature	(222) Grain Size (nm)	(222) d-Spacing (nm)	(440) d-Spacing (nm)	Lattice Constant a (nm)
110 °C	24.01	0.2921	0.2529	1.01188
150 °C	23.63	0.2922	0.2530	1.01236
190 °C	23.82	0.2921	0.2529	1.01193
230 °C	24.51	0.2921	0.2529	1.0

From Figure 5, it can be seen that after annealing at 230 °C for 1 h, the ITO films deposited at substrate temperatures of 110 °C and 150 °C still predominantly exhibit the (222) plane, while the diffraction peak intensities of planes such as (211), (440), and (400) also gradually increase. Compared to ITO films prepared at substrate temperatures of 190 °C and 230 °C, the difference is not significant; all show a crystalline state after annealing. Data from Table 3 show that after annealing, the grain size of the (222) plane for ITO prepared at different substrate temperatures significantly increases, and the lattice constant also gradually approaches the size of standard In_2O_3 (~1.0118 nm).

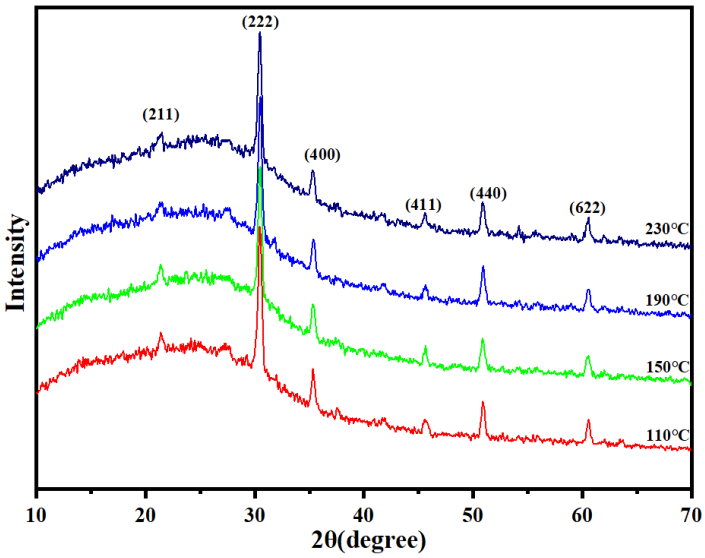


Figure 5. XRD patterns of ITO films (prepared at different substrate temperatures) after annealing at 230 °C in air for 1 h.

XRD analysis reveals the temperature dependence of film crystallization behavior. Before annealing, the sample deposited at 110 °C only shows a weak (222) diffraction peak, indicating the film is in a nanocrystalline state; the sample deposited at 150 °C begins to show (400) and (440) diffraction peaks, presenting mixed crystallization characteristics; when the temperature rises above 190 °C, the film is fully crystalline, and the preferred orientation changes from (222) to (400). After annealing at 230 °C, all samples show good crystalline states, but the low-temperature deposited samples have a more uniform grain size distribution.

Calculating grain size based on the Scherrer formula found that the grain size before annealing increases with deposition temperature, but after annealing treatment, the final grain sizes of samples deposited at different temperatures tend to be consistent (~32 nm). This phenomenon indicates that the amorphous films deposited at low temperatures undergo more significant structural reorganization during the annealing process, resulting in a more uniform microstructure.

3.2. Surface Morphology

ITO Film Surface Morphology and Roughness Analysis

Figure 6 shows the AFM microtopography of ITO films deposited at substrate temperatures of 150 °C, 190 °C, and 230 °C before and after annealing at 230 °C for 60 min. As evident from the figure, significant differences exist between the microstructures of the annealed (Figure 6a–c) and non-annealed (Figure 6d–f) ITO films. The annealed ITO films exhibit a uniform granular morphology, whereas the non-annealed films show block-like particles with randomly oriented grains.

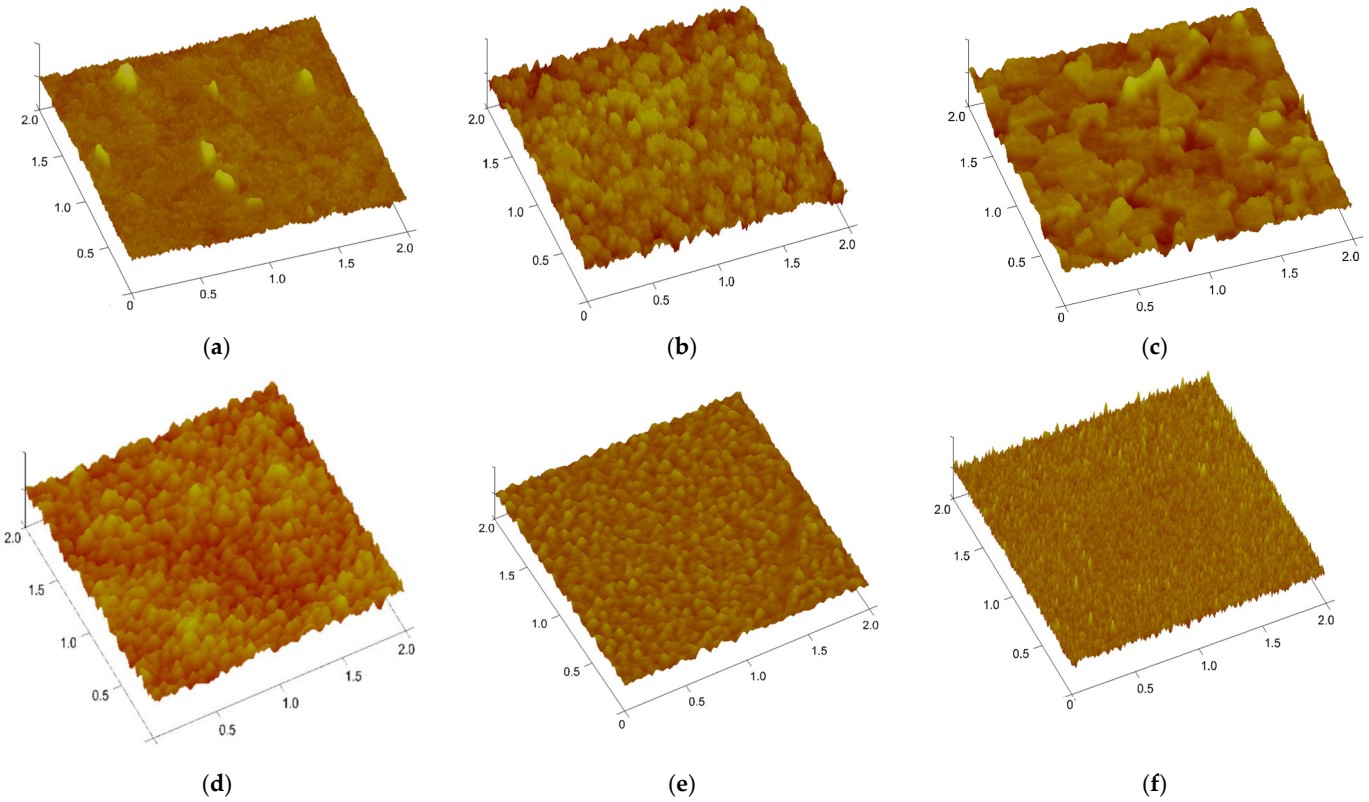


Figure 6. AFM images of ITO films as-deposited substrate temperatures of (**a**) 150 °C, (**b**) 190 °C, (**c**) 230 °C, and 110 °C substrate temperatures of ITO films annealing with (**d**) 150 °C, (**e**) 190 °C, (**f**) 230 °C.

As the substrate temperature increases prior to annealing, the surface roughness (Ra) of the post-annealed ITO films gradually decreases, recording values of 1.24 nm, 0.756 nm, and 0.649 nm, respectively. It is worth noting that the grain sizes calculated from XRD data differ from the AFM observations. Studies by Z. Ghorannevis et al. suggest that grain height significantly influences the surface roughness of ITO films [17], attributing this effect to surface protrusions and step edges. Grain height, defined by the height of these protrusions and steps, is determined by the deposition and crystallization conditions during crystal growth. Therefore, although the grain size varies for ITO films prepared at different substrate temperatures after annealing, the similar crystal growth conditions on the surface lead to a consistent trend in roughness reduction with increasing substrate temperature. Higher substrate temperatures promote improved crystal structure and enhanced surface flatness during growth, thereby reducing surface roughness.

In addition to grain height, other factors such as annealing temperature, redox properties, and atmospheric conditions also affect the surface roughness of ITO films. Higher annealing temperatures facilitate crystallization and grain growth, consequently reducing roughness. Variations in redox conditions and annealing atmosphere—such as oxidative

atmospheres removing surface defects and oxides, or reductive atmospheres reducing oxides and defects while promoting crystal growth and lattice refinement—also contribute to smoother surfaces.

Despite the lower surface roughness achieved with high-temperature deposition, the electrical performance of these films is comparatively inferior. This indicates that on flexible CPI substrates, surface roughness is not the dominant factor influencing electrical properties; instead, the deposition conditions play a more critical role.

3.3. Optical Properties

Transmittance tests (Figure 7) indicate that the visible light transmittance of all annealed samples at 550 nm was higher than 85%, with the sample deposited at 120 °C reaching 89.2%.

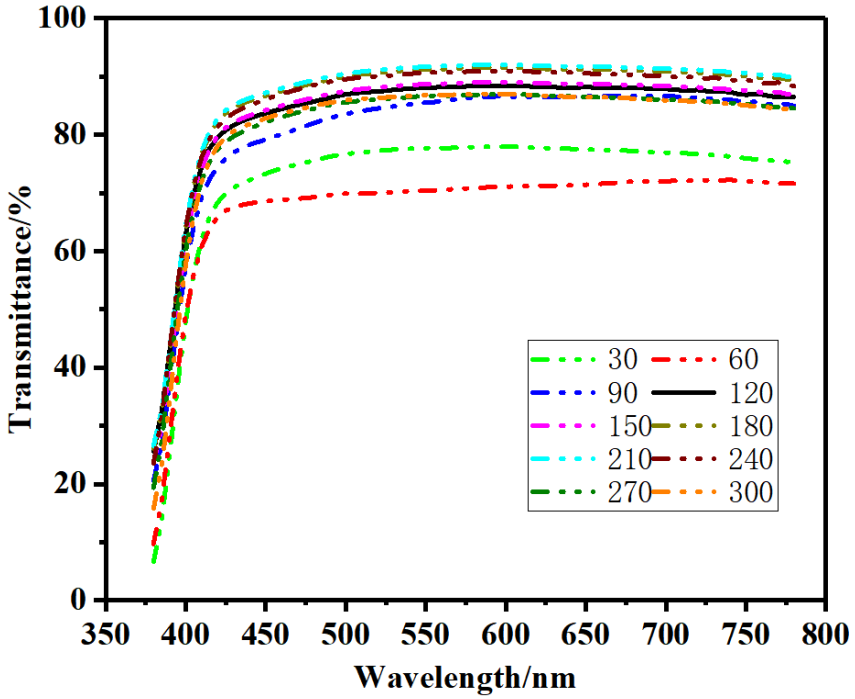


Figure 7. Transmittance Curve of Post-Annealing ITO vs. Deposition Temperature.

3.4. Electrical Properties

3.4.1. Resistivity Analysis

Figure 8 shows the variation in ITO film resistivity with deposition temperature after annealing. A non-monotonic trend was observed: resistivity decreased sharply from ~6000 μΩ·cm at 30 °C to a minimum of 203 μΩ·cm at 110 °C. However, as the temperature increased further above 150 °C, the resistivity increased slightly. This phenomenon can be attributed to the competition between increased grain size (reducing grain boundary scattering) and potential dopant deactivation or excessive reduction of oxygen vacancies at higher deposition temperatures, which can lower the effective carrier concentration.

The key electrical parameters for samples deposited at critical temperatures are summarized in Table 4. The sample deposited at 110 °C achieved the optimal combination of a high carrier concentration (8.7×10^{20} cm^{-3}) and a high mobility (35.2 $cm^2/V{\cdot}s$), resulting in the lowest resistivity. In contrast, the sample deposited at 230 °C, while having a comparable mobility, suffered from a significantly lower carrier concentration, leading to higher resistivity.

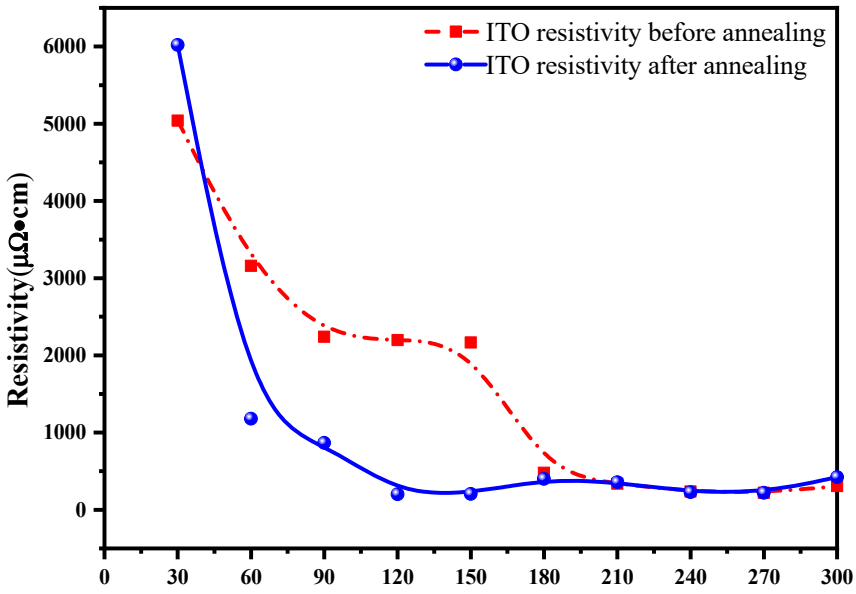


Figure 8. Curves of ITO resistivity vs. deposition temperature before and after annealing.

Table 4. Electrical parameters of ITO films after annealing at different deposition temperatures.

Substrate Temp.	Thickness (nm)	Resistivity (μΩ·cm)	Carrier Concentration (cm^{-3})	Mobility ($cm^2/V·s$)
110 °C	30 ± 2	203	8.7×10^{20}	35.2
150 °C	30 ± 2	225	7.9×10^{20}	35.1
230 °C	30 ± 2	385	4.8×10^{20}	33.8

3.4.2. Chemical State and Doping Mechanism

To reveal the chemical state evolution and doping efficiency mechanism of the annealed ITO films, high-resolution X-ray photoelectron spectroscopy (HR-XPS) was used to characterize the ITO film deposited at "110 °C + 230 °C atmospheric annealing for 60 min". Peak fitting using a Lorentz–Gauss mixed function was performed to analyze the chemical valence states and electronic structure of In, Sn, and O elements (Figure 9). Key analyses are as follows:

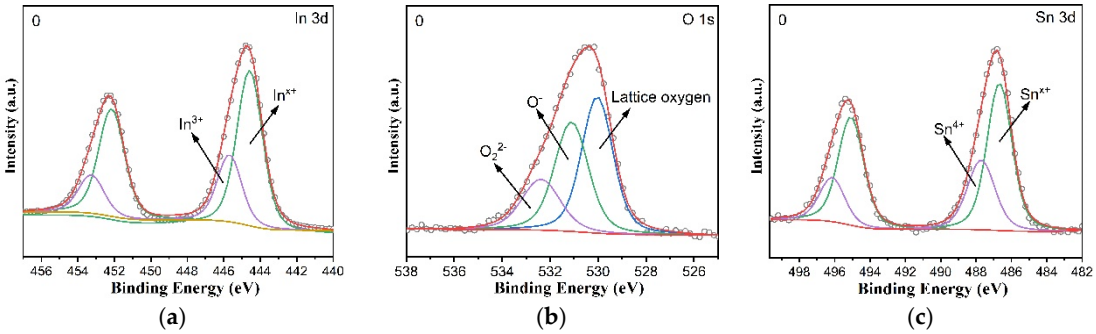


Figure 9. XPS valence state analysis of ITO thin films annealed at 230 °C for 60 min.

In 3d Orbital Chemical State

The In 3d orbital shows typical spin–orbit splitting features, split into In $3d_5/_2$ (low binding energy side) and In $3d_3/_2$ (high binding energy side), with a peak separation of approximately 3.3 eV (consistent with the standard spin-splitting spacing for In^{3+}). The characteristic peak of In $3d_5/_2$ is located at 444.6 eV, and In $3d_3/_2$ is at 452.0 eV, both

corresponding to the characteristic binding energy of In^{3+} in the In_2O_3 lattice. No peaks for In^0 (~441.5 eV) or other low-valent In ions were observed, indicating that In exists solely as In^{3+} in the film and no reduction reaction occurred; only a weak asymmetric broadening was observed on the high binding energy side of In $3d_5/_2$ (445.0–446.0 eV, marked as In_x^+), attributed to weak interaction between surface In^{3+} and adsorbed oxygen, belonging to surface slightly oxidized non-lattice In, whose impact on overall electrical performance is negligible.

Sn 3d Orbital Chemical State

The Sn 3d orbital also shows significant spin–orbit splitting, with a peak separation between Sn $3d_5/_2$ and Sn $3d_3/_2$ of approximately 8.4 eV (consistent with Sn element spin-splitting rules). Peak fitting results show two chemical states of Sn in the film:

(1) Effective doping state Sn^{4+}: The Sn $3d_5/_2$ characteristic peak is located at 486.7 eV, and Sn $3d_3/_2$ is at 495.1 eV, corresponding to the doping configuration where Sn^{4+} substitutes for In^{3+} in the In_2O_3 lattice (Sn^{4+}:In^{3+}). Each Sn^{4+} can provide one free electron to the conduction band, making it the core source of carriers in ITO films. Its relative content directly determines the doping efficiency.

(2) Ineffective doping state Sn^{2+}: A clear shoulder peak appears on the low binding energy side of Sn $3d_5/_2$ (485.2 eV), corresponding to the characteristic binding energy of Sn^{2+} (Sn $3d_3/_2$ is at 493.6 eV). The ionic radius of Sn^{2+} (0.93 Å) is much larger than that of In^{3+} (0.79 Å), making it unable to effectively substitute for In^{3+} and enter the lattice. It only exists in interstitial states or surface adsorption states, not only failing to contribute carriers but also potentially acting as carrier scattering centers, reducing carrier mobility.

Peak integration results show that the relative content ratio of Sn^{4+}/Sn^{2+} after annealing reaches 8.2:1, indicating that 230 °C atmospheric annealing significantly promotes the oxidative conversion of Sn^{2+} to Sn^{4+}, greatly improving doping efficiency—this directly correlates with the Hall test result of "carrier concentration reaching 8.7×10^{20} cm^{-3} for the 110 °C deposited sample".

O 1s Orbital Chemical State

The O_{1s} orbital can be fitted into three types of oxygen species, corresponding to different chemical environments. Their binding energies and physical meanings are as follows:

(1) Lattice oxygen ($O_{_L}$): Binding energy located at 530.0 eV, corresponding to O^{2-} strongly coordinated with In^{3+}/Sn^{4+} in the In_2O_3 lattice (In-O-Sn/In-O-In bonds). It is a marker of the film's crystal structure integrity; the higher its relative content, the better the lattice order.

(2) Oxygen vacancy-related oxygen ($O_{_V}$): Binding energy located at 531.5 eV, corresponding to low-coordination O atoms around oxygen vacancies. Oxygen vacancies are another source of carriers in ITO films (each oxygen vacancy can provide 2 free electrons), but their excessive presence can lead to lattice distortion and increased carrier scattering.

(3) Surface adsorbed oxygen ($O_{_ads}$):Binding energy located at 532.8 eV, corresponding to OH^-, H_2O, or O_2 molecules adsorbed on the film surface. It is unrelated to bulk electrical performance and only reflects the surface environment.

Peak integration results show that the relative content ratio of $O_{_L}/O_{_V}$ after annealing is 6.5:1, indicating that the excessive oxygen vacancies introduced by low-temperature oxygen-free deposition (Ar/O_2 = 120:0) are moderately suppressed during 230 °C atmospheric annealing ($O_{_L}$ proportion increases), forming a "controlled oxygen vacancy" state—retaining a certain number of carrier sources while avoiding scattering enhancement caused by lattice distortion, ultimately achieving an improvement in carrier mobility (35.2 $cm^2/V{\cdot}s$).

Comprehensive HR-XPS analysis reveals that the essence of the synergistic effect of "low-temperature deposition–post-annealing" is the precise regulation of chemical states—low-temperature oxygen-free deposition constructs an initial structure of "nanocrystalline + oxygen vacancies + Sn^{2+}", and subsequent 230 °C atmospheric annealing achieves two key transformations: ① Oxidation-driven $Sn^{2+} \rightarrow Sn^{4+}$, increasing effective doping concentration; ② Moderate oxidation reduces excessive oxygen vacancies, optimizing lattice order. Under the synergistic action of both, carrier concentration (n) and mobility (μ) are simultaneously improved, ultimately achieving a significant reduction in film resistivity (203 μΩ·cm).

3.5. Bending Resistance

As shown in Table 5 above, the bending test results show that the substrate temperature significantly affects the $\Delta R/R_0$ change rate of CPI-based ITO films. At a substrate temperature of 30 °C, after 500 bending cycles, the initial sheet resistance change rate was already 16.4%, losing application value. At a substrate temperature of 90 °C, $\Delta R/R_0$ increased rapidly with the number of bending cycles, sharply rising to 41.7% at 4000 cycles. At 120 °C and 150 °C, after 100,000 bending cycles, the $\Delta R/R_0$ change rates were 7.52% and 7.81%, respectively, possessing commercial use value. However, when the substrate temperature further increased above 210 °C, the bending resistance decreased somewhat. Therefore, the optimal substrate temperature for ITO films with the best bending endurance is 120 °C. The relationship between the number of bending cycles and the substrate temperature of the ITO thin film is shown in Figure 10 below.

Table 5. Bending Test Data of ITO Thin Films at Different Temperatures.

Sub. Temper	**30 °C**	**90 °C**	**120 °C**	**150 °C**	**210 °C**	**240 °C**	**300 °C**
Bending Cycles	**$\Delta R/R_0$ (%)**	**$\Delta R/R_0$ (%)**	**$\Delta R/R_0$ (%)**	**$\Delta R/R_0$ (%)**	**$\Delta R/R_0$ (%)**	**$\Delta R/R_0$ (%)**	**$\Delta R/R_0$ (%)**
500	0.164	0.05	0	0	0	0	0.18
1000	0.1863	0.0889	0.01	0.018	0.02	0.02	0.3245
2000	0.2412	0.0938	0.0125	0.0142	0.023	0.023	0.4312
4000	0.7874	0.4171	0.0189	0.0197	0.025	0.025	0.5721
8000	0.9627	0.4985	0.0235	0.0236	0.027	0.027	0.637
16,000	1	0.6136	0.0273	0.0256	0.03	0.03	0.934
32,000	1	0.76	0.0325	0.0315	0.033	0.033	1
64,000	1	0.841	0.0572	0.058	0.061	0.061	1
100,000	1	0.988	0.0752	0.078	0.088	0.088	1

Some studies point out that stress concentration at defect points initiating cracks and delamination forming wrinkles are key factors leading to the decline of ITO conductivity [18]. During ITO film bending, due to stress concentration, microcracks first generate at defect points and extend along both ends in the direction of the bending roller axis, forming transverse cracks. As the number of bending cycles increases, the number of cracks increases, and the ITO at the crack surface peels off and sheds in fragments [19]. Cairns et al. pointed out that there is a corresponding relationship between the number of cracks per unit length n_d and the bending strain ε; n_d increases with the growth of ε, but n_d does not increase indefinitely, eventually tending towards a saturation value [20]. The number of cracks gradually stabilizes with the increase in bending cycles. Because the deeper ITO experiences less deformation and remains partially connected, retaining some conductive channels, the resistance change rate curve shows rapid growth initially, then tends to a relatively flat high-resistance state. The damaged ITO fragments at the crack sites suffer increased damage during repeated bending. Continuing to increase the number

of bends will cause the ITO at the cracks to fall off, blocking the conductive channels and losing conductivity.

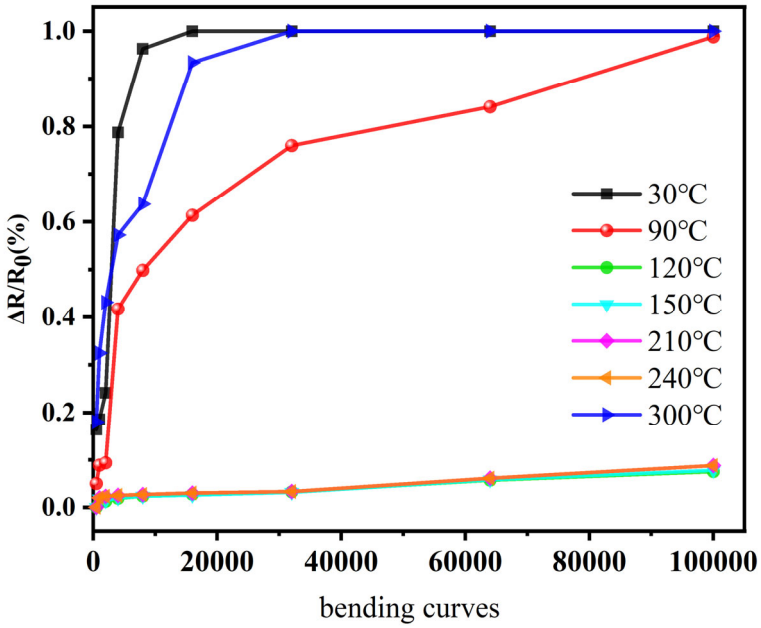


Figure 10. Bending Cycles vs. Substrate Temperature Curve of ITO films on CPI.

4. Conclusions

Using a DC magnetron sputtering method combined with a "low-temperature deposition (110 °C)–230 °C atmospheric annealing" synergistic strategy, high-performance flexible ITO conductive films were successfully prepared under conditions far below the CPI tolerance limit. The optimal sample exhibited excellent comprehensive performance: low resistivity (203 μΩ·cm), high visible light transmittance (89.2%), low surface roughness (0.76 nm), and outstanding bending endurance ($\Delta R/R_0 < 10\%$ after 100,000 cycles at R = 5 mm).

The study revealed that the substrate temperature critically regulates the initial crystallization state. Samples deposited at 110 °C have a nanocrystalline structure, while those deposited above 150 °C gradually develop a (400) preferred orientation. After annealing, all samples achieved good crystallization, but the low-temperature deposited nanocrystalline films underwent more sufficient structural reorganization, resulting in a uniform final grain size (~24 nm) and optimal crystal integrity.

The core role of the annealing process is the precise regulation of chemical states: the complete oxidation of Sn^{2+} to Sn^{4+} (Sn^{4+}/Sn^{2+} = 8.2:1) increases the effective doping concentration, and the controlled adjustment of oxygen vacancies (O_L/O_V = 6.5:1) optimizes lattice order. This synergy simultaneously enhances carrier concentration (8.7×10^{20} cm^{-3}) and mobility (35.2 $cm^2/V{\cdot}s$).

While higher deposition temperatures led to lower surface roughness after annealing, the electrical performance deteriorated, confirming that crystal quality and doping efficiency are the dominant factors for the electrical performance of flexible CPI-based ITO films, with surface roughness being a secondary factor.

This work elucidates the crystallization kinetics and doping mechanisms of ITO films under low-temperature conditions, breaking through the contradiction between "flexible substrate heat resistance–ITO performance". It provides a viable, industrially feasible low-temperature technical pathway for preparing high-performance transparent electrodes in flexible vehicle-mounted touch displays and wearable electronics.

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